Sinie van der Ben

linkedin.com/in/sinievanderben · Zurich, Switzerland

Research Profile

PhD researcher at ETH Zurich (D-INFK), working interpretability of large language models, making model internals understandable for humans, and testing whether the resulting measurements can be trusted. Four 2026 papers spanning visualization, interpretability and NLP. Python and PyTorch engineer with prior industry ML and cloud experience. (#human-ai-vis)

Education

PhD Computer Science — Interpretability & Visualization of LLMs Nov 2024 – Present
ETH Zurich, D-INFK, IVIA Lab · Advisor: Prof. Mennatallah El-Assady
Zurich, Switzerland

- Showed that LLM-as-a-Judge scores for sparse autoencoder features are dominated by pipeline choices
- Current projects span investigating cheaper steering methods and testing proposed steering methods for abstract concepts
- Host a monthly Interpretability × Apertus standup at the ETH AI Center; supervise MSc and BSc research projects

MSc Artificial Intelligence — With Distinction (GPA 8.7/10 – 4/4) Sep 2020 – Jul 2023
Utrecht University · Thesis conducted at ETH Zurich

- Thesis: "Explainability for Network Anomaly Detection" (grade 8.7/10), quantitative XAI evaluation for anomaly detection systems
- Descartes Honours College, a selective programme for top students across disciplines

BSc Artificial Intelligence (GPA 8.0/10) Sep 2016 – Apr 2020
Utrecht University

Publications & Research Output

- Yan, X., Sevastjanova, R., **van der Ben, S.**, El-Assady, M., & Wang, B. (2026). Explainable Mapper: Charting LLM Embedding Spaces Using Perturbation-Based Explanation and Verification Agents. *IEEE Transactions on Visualization and Computer Graphics (Proc. IEEE VIS 2026)*
- **van der Ben, S.**, Baur, R., Metz, Y., & El-Assady, M. (2026). Where Do Models Find Happiness? Emotion Vectors in Open-Source LLMs. *ICML 2026 Mechanistic Interpretability Workshop (virtual poster)*
- **van der Ben, S.**, Roch, N., Hedström, A., & El-Assady, M. (2026). Building Fast, Evaluating Slow: Pipeline Choices Dominate Autointerpretability Score Variance. *ICML 2026 Mechanistic Interpretability Workshop (poster)*
- Jeong, D., Lee, Y., Choi, E., **van der Ben, S.**, & Kim, J. (2026). Investigating Assistant Bias in LLM User Simulators Using a Role Vector. *EMNLP 2026*

Open Source, Community & Teaching

- TransformerLens: added support for Apertus (Swiss AI open LLM), making it usable with standard interpretability tooling (pull request).
- Research blog post on Apertus Claritas explaining the emotion-vectors paper (blog post).
- Teaching Assistant, Large-Scale AI Engineering, ETH Zurich, FS 2026 (course)

Research & Industry Experience

Research Assistant Oct 2023 – Nov 2024
Utrecht University, Utrecht, the Netherlands, part-time

- Applied ML methods to dietary-pattern analysis and ultra-processed food classification
- Cross-institutional collaboration between Utrecht University, the Dutch National Institute for Public Health (RIVM), and Harvard Medical School

Machine Learning Engineer Nov 2023 – Sep 2024
Enjins, Utrecht, the Netherlands

- Designed and deployed ML pipelines and data-platform infrastructure (Python, Terraform, Kubernetes, AWS EKS, Azure)
- Built an Azure data platform from the ground up; contributed to data-integration proposals and cloud architecture

Working Student — Risk Advisory & Cloud Engineering Nov 2022 – Oct 2023
Deloitte, Amsterdam, the Netherlands

- Built serverless cloud tooling for the Risk Advisory team (Go, Terraform, AWS)

Deep Learning Intern — Multimodal Deepfake Detection Feb 2022 – Aug 2022
DuckDuckGoose AI, Delft, the Netherlands

- Implemented a dual-stream multimodal deepfake detector: separate vision and language encoders fused through intermediate cross-modal attention layers

Student Assistant — PROVEE Project Jan 2021 – Dec 2021
Utrecht University

- Built incremental PCA-based visualisation of high-dimensional ML embeddings for PROVEE (Progressive Explainable Embeddings), letting users watch an embedding space form rather than only see its final projection
- Wrote two Medium posts on visualising large embedding spaces (first, second).

Deep Learning Intern — Cancer Cell Detection Feb 2021 – Jul 2021
Prinses Máxima Centrum, Utrecht, the Netherlands

- Used GANs to augment training data for cancer cell detection with limited fluorescent markers, improving classifier performance in low-data regimes

Hackathons

ETH AI Center × Anthropic Sprint Jun 2026

- Built a real-time tool that listens to a group conversation and lets a chart emerge from the discussion: speaker-attributed transcription drives continuously regenerated, model-written Python rendered in Anthropic's code-execution sandbox

Future of Health Hackathon May 2026

- Built an interactive graph view giving pathologists an instant visual overview of patient test results across heterogeneous testing profiles

ETH AI Club Hackathon (short) Oct 2026

- Test strength of Apertus v1.5 with special token-injevtion and attacks on thinking trace.

Technical Skills

Core: Python (advanced), PyTorch, TransformerLens, JavaScript/D3 (basic), C++ (basic), Go (basic)
ML: LLM interpretability (SAEs, probing, activation steering), interactive visualization of model internals, XAI evaluation, multimodal deep learning
Infrastructure: Terraform, Kubernetes, AWS EKS, Azure, Docker
Languages: Dutch (native), English (fluent), German (intermediate)

Other

Competitive Rower Nov 2018 – Jun 2023

- Trained 7–10 sessions per week at national competitive level; elected Career Days secretary, organising a week of speaker events sponsored by Philips